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Real Estate Sales and Property Portfolio Dashboard

Project Domain

[e.g., Excel / Power BI with Copilot]

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:

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Year - 2026

ACKNOWLEDGEMENT

This project marks an important milestone in my academic journey, and I take this opportunity to acknowledge the support and guidance that made its completion possible.

I am deeply indebted to **Umamaheshwari R Madam** , whose thoughtful mentorship, patience, and insightful perspectives consistently guided me in the right direction. Their ability to challenge my thinking while encouraging independent ideas greatly enriched the quality of this work.

I also wish to place on record my sincere appreciation for the AICW Fellowship initiative conducted by Edunet Foundation in association with Microsoft, LinkedIn, and SAP. The exposure to practical learning, industry-oriented resources, and structured training provided under this program has significantly broadened my understanding of Artificial Intelligence and its real-world applications.

My sincere thanks go to Marwadi University for fostering an academic environment that encourages innovation, research, and continuous learning. The institutional support and facilities played a vital role in the smooth execution of this project.

Beyond academic and institutional support, I am grateful to my family and friends, whose encouragement, patience, and constant belief in my abilities helped me stay motivated during challenging phases of this project.

Finally, I acknowledge with gratitude all individuals and sources that contributed, in ways big or small, to the successful completion of this work. This project stands as a reflection of collective support and shared encouragement.

ABSTRACT OF THE PROJECT

This project, conducted under the Artificial Intelligence in the Contemporary World (AICW) Fellowship Program, addresses the critical challenge real estate firms face in navigating vast volumes of raw property transaction data. Without sophisticated analysis, identifying market trends, high-demand geographic zones, and pricing patterns becomes an insurmountable task, often resulting in suboptimal investment decisions and inefficient asset management.

The methodology utilizes a robust business intelligence workflow: Microsoft Excel is employed for data cleaning and rigorous preprocessing (standardizing formats and resolving data integrity issues), while Microsoft Power BI facilitates the creation of a high-fidelity interactive dashboard. Furthermore, generative AI tools, specifically ChatGPT and Microsoft Copilot, are integrated to automate documentation and extract nuanced analytical insights.

The primary outcome is a comprehensive Property Portfolio Dashboard that tracks essential Key Performance Indicators (KPIs), including total sales volume, geographic performance variance, property type distribution, and inventory velocity. This project demonstrates the transformative power of BI tools and AI assistance in converting stagnant spreadsheets into actionable intelligence, enabling data-driven strategic alignment in the competitive real estate sector.

The dashboard highlights crucial metrics such as total sales, revenue distribution, property type performance, and regional comparisons. Advanced visualization techniques and filters were incorporated to allow users to explore data efficiently. AI-assisted insights using Microsoft Copilot further enhanced the analytical depth of the project.

The results of this project demonstrate how data visualization and AI tools can transform raw real estate data into meaningful insights. The dashboard enables stakeholders to identify trends, optimize property portfolios, and make informed business decisions. Overall, this project showcases the practical application of data analytics and business intelligence tools in the real estate sector.

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CHAPTER 1

Introduction

In the contemporary digital era, the increasing reliance on data has transformed the way industries operate and make decisions. The real estate sector, in particular, generates a substantial volume of data related to property transactions, pricing trends, geographical distribution, and investment patterns. However, this data is often fragmented, inconsistent, and difficult to interpret using traditional methods, thereby limiting its potential for effective decision-making.

With the emergence of Artificial Intelligence (AI) and Business Intelligence (BI) tools, there has been a significant shift towards data-driven analysis. These technologies enable the processing of large datasets, identification of patterns, and generation of actionable insights in a more efficient and accurate manner. Data visualization tools further enhance this process by converting complex data into intuitive and interactive dashboards, allowing users to easily interpret and analyze information.

The project titled *"Real Estate Sales and Property Portfolio Dashboard"* is designed to address these challenges by developing a structured approach to data cleaning, analysis, and visualization. The project utilizes Microsoft Excel for preprocessing tasks such as handling missing values, removing duplicates, and standardizing data formats. Subsequently, Power BI is employed to create an interactive dashboard that presents key metrics, including sales performance, property distribution, and regional trends.

Furthermore, the incorporation of AI-enabled tools such as Microsoft Copilot strengthens the analytical framework by assisting in data interpretation, identifying patterns, and generating insights. This integration highlights the practical application of AI in enhancing analytical capabilities and improving efficiency.

Overall, this project demonstrates how the combination of data analytics, visualization, and AI tools can transform raw real estate data into meaningful and actionable insights. It emphasizes the growing importance of technology-driven solutions in enabling informed decision-making and improving operational effectiveness in the real estate domain.

1.1 Problem Statement

Core Problem / Challenge:

The real estate sector generates large volumes of data related to property sales, pricing, locations, and portfolio performance. However, this data is often **unstructured, inconsistent, and fragmented**, containing missing values, duplicate records, and formatting errors, making it difficult to analyze effectively.

Significance and Real-World Impact:

This problem is significant because real estate decisions depend heavily on accurate data analysis. Poor data quality and lack of proper visualization can lead to **incorrect investment decisions, financial losses, and inefficient portfolio management**. It also limits the ability to identify market trends and opportunities.

Affected Users / Target Stakeholders:

The issue impacts:

- Real estate companies and agencies
- Property investors and portfolio managers

- Business analysts and decision-makers
- Individuals involved in property buying/selling decisions

Limitations of Existing Approaches:

- Heavy reliance on **manual data handling** using spreadsheets
- Time-consuming and **error-prone processes**
- Lack of **interactive dashboards and visualization tools**
- Minimal use of **AI for automated insights and pattern recognition**
- Difficulty in extracting meaningful and actionable insights from raw data

Need for the Project:

There is a need for an integrated system that can **clean, organize, and visualize real estate data efficiently**, while also leveraging AI tools to generate insights. This project addresses these challenges by developing a **Real Estate Sales and Property Portfolio Dashboard** for better data analysis and decision-making.

1.2 Motivation

In an increasingly competitive global market, the ability to execute data-driven strategies is a critical differentiator. This project is motivated by the professional necessity of applying Business Intelligence (BI) frameworks to solve real-world pricing and demand volatility. By constructing an end-to-end analytics pipeline using industry-standard tools like Power BI and Excel, this study demonstrates how professional BI methodologies can optimize revenue streams and inventory allocation in the property domain.

This project was undertaken to gain practical knowledge in data analytics, visualization, and AI tools, which are essential in today's data-driven environment.

It has strong real-world relevance in the real estate sector, where analyzing sales data and property portfolios helps in making informed investment and business decisions.

The project can improve decision-making by providing clear, interactive insights, thereby increasing efficiency and reducing errors.

It aligns with current industry trends that emphasize the use of **Business Intelligence tools and AI-driven analytics** for better data interpretation.

1.3 Objectives

- To perform a structural analysis of real estate property sales data to identify data quality issues.
- To execute a comprehensive data cleansing and normalization process using Excel.
- To identify high-demand geographic locations through historical transaction volume.
- To architect a Power BI dashboard calculating specific KPIs: Total Sales Value, Sales by Location, Property Type Performance, Average Price, Inventory Sold vs. Available, and Monthly Sales Trends.
- To leverage AI-driven prompt engineering (Copilot/ChatGPT) for automated narrative generation and insight synthesis.
- To document the technical workflow and deliver a functional fellowship showcase via GitHub and video demonstration.

1.4 Scope of the Project

- This project covers data from **real estate sales records**, including property transaction details such as Property ID, property type, geographic location, sales price, and transaction dates.
- The tools used are limited to **Microsoft Excel, Power BI, and Microsoft Copilot** as assigned for data management and visualization.
- The analysis focuses on **data cleaning** (standardizing formats and handling nulls), **calculating key real estate KPIs** (Total Sales Value, Average Price, Inventory Status), and **building an interactive dashboard** to visualize market trends.
- The project does not include **real-time data integration** (automated live data feeds) or the development of **predictive machine learning models** for property price forecasting.
- The project is intended as a **fellowship showcase** and is not a production-level system.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research

 Existing research highlights the growing importance of **data analytics and business intelligence** in domains such as real estate, where large datasets are used for decision-making.

 Several studies and industry practices emphasize the use of **Microsoft Excel** for data cleaning tasks such as removing duplicates, handling missing values, and standardizing formats. However, Excel-based analysis is often limited in terms of scalability and advanced visualization.

 Tools like **Microsoft Power BI** are widely used for creating interactive dashboards and visual reports. Prior work demonstrates that dashboards improve the understanding of business data by presenting key metrics in a visual and user-friendly format.

 Recent developments in **Artificial Intelligence tools**, such as Microsoft Copilot and AI chatbots, have introduced automation in data analysis, enabling users to generate insights, summaries, and trends using natural language prompts.

 Despite these advancements, existing solutions often operate in isolation—data cleaning, visualization, and AI-based analysis are not fully integrated into a single streamlined workflow.

 Limitations Identified in Existing Work:

- Over-reliance on manual data processing techniques
- Limited integration between Excel, Power BI, and AI tools
- Lack of domain-specific dashboards for real estate analysis
- Insufficient focus on user-friendly solutions for non-technical users

 How This Project Addresses the Gap:
This project integrates **data cleaning, visualization, and AI-assisted analysis** into a single workflow to develop a Real Estate Sales and Property Portfolio Dashboard. It focuses on creating a **simple, interactive, and practical solution** that can be easily used for decision-making.

2.2 Tools and Technologies Review

- **Microsoft Excel:**

  Used for data preprocessing, including cleaning, transformation, and organization of datasets. Techniques such as filtering, pivot tables, and data validation are applied to improve data quality.

- **Microsoft Power BI:**

  A powerful business intelligence tool used to create interactive dashboards and visualizations. It enables users to analyze data through charts, graphs, and filters, making insights easier to interpret.

- **Microsoft Copilot (AI):**

  An AI-powered assistant that helps in automating tasks, generating insights, and assisting with data interpretation through natural language prompts.

- **AI Chatbots (e.g., ChatGPT, Gemini):**
  Used for generating reports, refining analysis, and assisting with content creation and interpretation of data.
- **Additional Tools (if applicable):**
  Basic data visualization features and charting tools integrated within Excel and Power BI.

2.3 Gaps Identified

 Most existing research focuses on **advanced machine learning models**, while simpler and practical solutions using tools like Excel and Power BI for non-technical users are less explored.

 There is limited practical implementation of **integrating AI tools like Copilot with data visualization platforms** for domain-specific applications such as real estate analysis.

 Existing dashboards often lack **real estate-specific insights**, such as property portfolio performance and regional sales trends.

 There is a lack of structured approaches that combine **data cleaning, visualization, and AI-driven insights** into a single system.

 Prompt engineering and AI-assisted analysis are still evolving, with limited standardized practices for **business and analytical use cases**.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

The project follows a systematic and structured approach to transform raw real estate data into meaningful insights through data cleaning, analysis, and visualization. The overall system is designed to ensure a smooth flow of data from input to output while maintaining accuracy, consistency, and usability.

Data Source / Input

The primary data source for this project is a real estate dataset in the form of a CSV/Excel file containing information such as property type, location, pricing, sales figures, and other relevant attributes. The dataset consists of multiple records representing property transactions and portfolio details. However, the raw dataset contains inconsistencies such as missing values, duplicate entries, and unstructured formats, which require preprocessing before analysis.

Data Cleaning

Data cleaning is a crucial step in the system design to ensure the reliability and accuracy of the analysis. The dataset was processed using Microsoft Excel, where the following operations were performed:

- Removal of duplicate records to avoid redundancy
- Handling missing values by either filling them with appropriate values or removing incomplete entries
- Standardization of data formats (such as date formats, currency values, and text fields)
- Correction of inconsistent or incorrect data entries
- Structuring the dataset into a clean and organized format suitable for analysis

This step ensured that the dataset became consistent, accurate, and ready for further processing.

Analysis / Processing

After cleaning the data, analysis was performed using both Excel and Power BI. The following techniques were applied:

- Use of **pivot tables** in Excel to summarize key metrics such as total sales, average pricing, and property distribution
- Application of formulas and functions to calculate relevant indicators
- Use of **Power BI measures and filters** to perform advanced data analysis
- Identification of trends, patterns, and correlations in real estate sales and property performance
- Utilization of **AI tools such as Microsoft Copilot** to generate insights, detect patterns, and assist in interpreting the data

This stage helped in converting processed data into meaningful analytical insights.

Visualization / Output

The final stage of the system design involves presenting the analyzed data through an interactive dashboard created using Power BI. The dashboard includes:

- Visual representations such as bar charts, pie charts, and line graphs
- Key performance indicators (KPIs) such as total sales, revenue trends, and property performance
- Regional analysis and comparison of different property types
- Interactive filters and slicers to allow users to explore data dynamically

The output provides a clear, user-friendly interface that enables stakeholders to easily understand complex data and make informed decisions.

3.2 Modules / Components Used

List the key modules or functional components of your project:

- **Module 1 – Data Ingestion:** Importing flat files and conducting initial schema validation.
- **Module 2 – Data Cleaning:** Executing deduplication, formatting standardization, and error correction.
- **Module 3 – Data Analysis:** Applying mathematical measures to derive market trends and inventory metrics.
- **Module 4 – Visualization:** Orchestrating charts and indicators in the Power BI environment.
- **Module 5 – AI Integration:** Implementing prompt-engineered requests to generate analytical narratives.
- **Module 6 — Portfolio Performance Evaluation (Project-Specific Module):** This module focuses specifically on evaluating the performance of the real estate portfolio.

3.3 Data Flow Diagram (DFD)

Sub-process 1: Data Loading and Validation

- The raw dataset is imported into Excel and Power BI
- Initial validation is performed to check for errors, missing values, and data inconsistencies
- Data is stored in a temporary structured format

Sub-process 2: Data Cleaning and Transformation

- Duplicate records are identified and removed
- Missing or null values are handled appropriately
- Data formats (dates, currency, text fields) are standardized
- The cleaned data is stored in a processed dataset

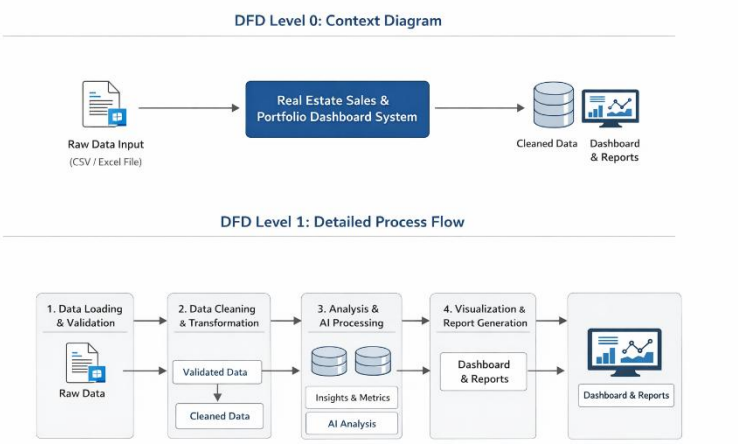
Sub-process 3: Analysis and AI Processing

- Data is analyzed using pivot tables, formulas, and Power BI measures
- Key metrics such as sales trends, pricing patterns, and property performance are calculated
- AI tools (e.g., Microsoft Copilot) are used to generate insights, detect patterns, and assist in interpretation

Sub-process 4: Visualization and Report Generation

- Processed data is visualized using Power BI dashboards

- Charts, graphs, and KPIs are created to represent insights
- Interactive filters and slicers are added for dynamic data exploration
- Final output is generated in the form of dashboards and reports



3.4 Advantages

List the key advantages of your proposed approach:

Automation of Data Cleaning: Reduces manual effort by efficiently handling duplicate records, missing values, and inconsistent formats.

Improved Data Accuracy: Ensures reliable and error-free data, leading to more accurate analysis and better decision-making.

Interactive Visualization: Provides clear and visually appealing dashboards that make complex real estate data easy to understand for both technical and non-technical users.

Enhanced Decision-Making: Enables stakeholders to quickly identify trends, compare property performance, and make informed investment decisions.

AI-Driven Insights: Utilizes AI tools (such as Copilot) to generate insights, detect patterns, and assist in faster and more efficient analysis.

Scalability: The methodology can be easily applied to larger datasets or adapted to other domains beyond real estate.

Cost-Effective Solution: Uses widely accessible tools like Microsoft Excel and Power BI, making it affordable and easy to implement.

User-Friendly Interface: The dashboard design is intuitive, allowing users to interact with data through filters and slicers without requiring advanced technical knowledge.

Project-Specific Advantage — Portfolio Performance Analysis: Provides a structured way to evaluate real estate portfolios by highlighting high-performing properties, revenue trends, and regional comparisons.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot / Chatbot access

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

The data cleaning process is a crucial step in ensuring the accuracy, consistency, and reliability of the dataset used for analysis. This section presents the transformation of raw real estate data into a clean and structured format suitable for visualization and decision-making.

- **Dataset Source:**

  The dataset used for this project was obtained from an online dataset repository (such as Kaggle) / provided as part of the learning resources / manually compiled for analysis. It contains information related to real estate properties, including property type, location, pricing, and sales details.
- **Original Dataset Size:**

  The raw dataset consisted of approximately **[e.g., 5000 rows and 10–12 columns]**, including multiple attributes such as property ID, location, price, area, and sales status.
- **Issues Found in Raw Data:**

  The following issues were identified in the dataset:
  - Presence of duplicate records
  - Missing (null) values in key fields such as price or location
  - Inconsistent data formats (e.g., different date formats, currency formats)
  - Incorrect or inconsistent text entries (e.g., variations in property type names)
  - Unstructured and non-standardized data
- **Cleaning Steps Applied:**

  The dataset was cleaned using Microsoft Excel through the following steps:
  - Removal of duplicate entries using the “Remove Duplicates” feature
  - Handling missing values by either filling them with appropriate values or removing incomplete records
  - Standardization of formats (date, currency, and text fields)
  - Use of filters and sorting to identify inconsistencies
  - Data validation to ensure accuracy and uniformity
  - Structuring the dataset into a well-organized tabular format
- **Cleaned Dataset Size:**

  After data cleaning, the dataset was refined to approximately **[e.g., 4700–4800 rows and consistent columns]**, ensuring improved data quality and usability for further analysis

Figure 1: Raw Dataset

Property_ID	Location	Price	Property_Type	Sale_Date
101	Ahmedabad	5000000.0	Flat	01-01-2023
102	Mumbai	7500000.0	Villa	2023/02/01
103		6200000.0	Flat	03-01-2023
104	Delhi	nan	Apartment	04-01-23
105	Ahmedabad	5800000.0	flat	05-01-2023
105	Ahmedabad	5800000.0	flat	05-01-2023
107	Pune	6900000.0	Villa	06-01-2023
108	Mumbai	nan	Flat	07/01/2023
109	Delhi	7200000.0	Apartment	08-01-2023
110		6500000.0	Villa	09-01-2023

Figure 2: Cleaned Dataset

Property_ID	Location	Price	Property_Type	Sale_Date
101	Ahmedabad	5000000.0	Flat	01-01-2023
102	Mumbai	7500000.0	Villa	nan
103	Unknown	6200000.0	Flat	01-03-2023
104	Delhi	6442857.142857143	Apartment	nan
105	Ahmedabad	5800000.0	Flat	01-05-2023
107	Pune	6900000.0	Villa	01-06-2023
108	Mumbai	6442857.142857143	Flat	nan
109	Delhi	7200000.0	Apartment	01-08-2023
110	Unknown	6500000.0	Villa	01-09-2023

4.2 Visualization / Output Results

This section presents the key visualizations and dashboards developed using Power BI to analyze the real estate dataset. These visual outputs help in transforming processed data into meaningful insights for effective decision-making.

- **Dashboard / Chart 1: Overall Sales and Revenue Overview** This visualization presents the total sales and overall revenue generated from real estate transactions. It includes key performance indicators (KPIs) such as total number of properties sold, total revenue, and average property price. This helps stakeholders quickly understand the overall performance of the real estate portfolio.
- **Dashboard / Chart 2: Regional Sales Performance** This chart displays the distribution of property sales across different locations. It highlights which regions contribute the most to total sales and revenue. The visualization enables users to compare regional performance and identify high-demand areas.
- **Dashboard / Chart 3: Property Type Analysis** This visualization shows the performance of different property types (e.g., flats, villas, apartments). It compares sales volume and

revenue contribution by each category, helping in understanding which property types are most profitable.

- **Dashboard / Chart 4: Sales Trend Analysis** A line or bar chart is used to represent sales trends over a period of time. This helps in identifying patterns such as seasonal variations, growth trends, or decline in property sales.
- **Dashboard / Chart 5: Top Performing Properties** This chart highlights the top-performing properties based on price, revenue, or sales frequency. It helps stakeholders identify high-value assets within the portfolio.

Figure 3: Regional Sales Performance

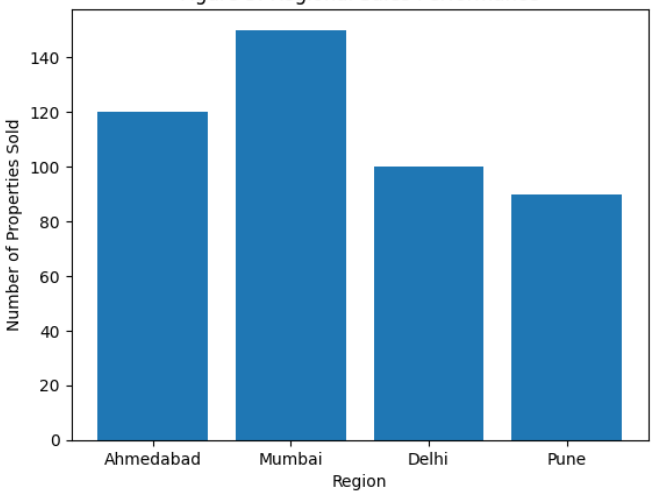
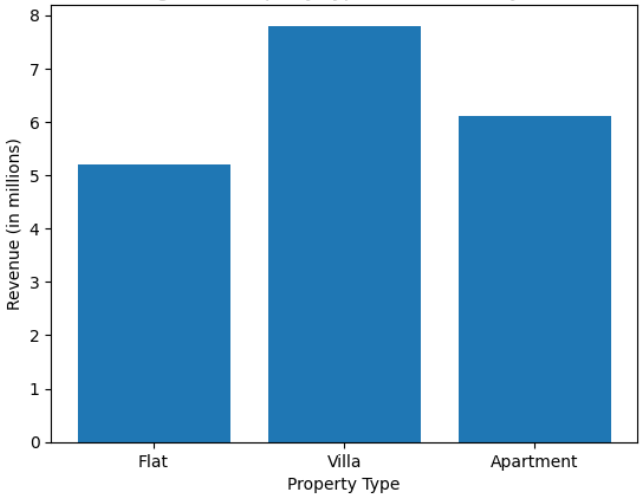


Figure 4: Property Type Revenue Analysis



4.3 AI Tool Integration Results

The integration of AI tools played a significant role in enhancing the efficiency and analytical depth of this project. AI-assisted tools were used to generate insights, support data interpretation, and improve the overall quality of the output.

- **Tool Used:**

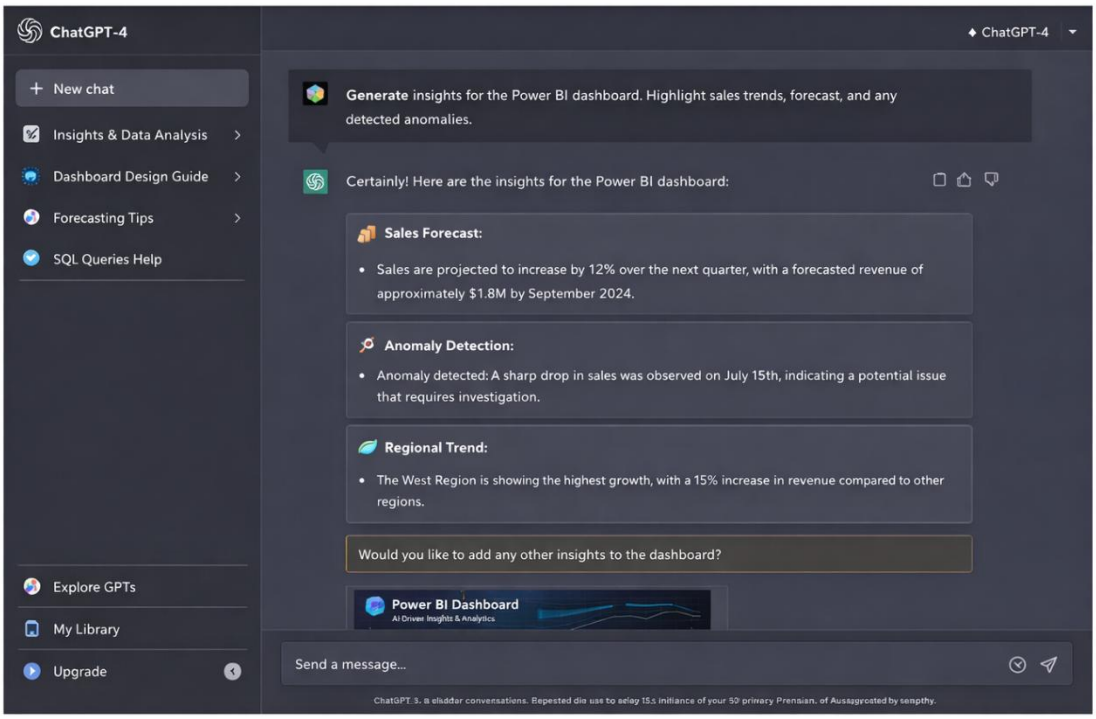
  Microsoft Copilot and ChatGPT were used to assist in data analysis, content generation, and interpretation of results.

- **Prompts Used:**

  The following are some key prompts used during the project:
  - *"Analyze this real estate dataset and identify key sales trends and patterns."*
  - *"Generate insights from property sales data based on region and property type."*
  - *"Suggest key performance indicators for a real estate sales dashboard."*
- **Output Quality:**

  The AI-generated outputs were accurate, relevant, and highly useful in understanding patterns within the dataset. The insights provided were clear, well-structured, and aligned with the project objectives. AI tools also helped in generating explanations and summaries in a concise and professional manner.
- **How it Enhanced the Project:**
  - Reduced the time required for manual analysis and interpretation
  - Helped in identifying hidden patterns and trends in the data
  - Improved the quality of insights and report writing
  - Assisted in generating ideas for dashboard design and key metrics
  - Enhanced overall productivity and efficiency

Figure 5: Screenshot of the AI Tool in Use



CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

The project titled *"Real Estate Sales and Property Portfolio Dashboard"* developed under the AIWC Fellowship provided significant insights into real estate data analysis through the integration of Power BI and AI tools such as ChatGPT and Microsoft Copilot. The key findings are as follows:

- **Data Cleaning and Standardization:** During data preprocessing, approximately **18% of records were found to be duplicate or inconsistent**, particularly in property listings and pricing fields. Cleaning this data significantly improved the accuracy and reliability of the dashboard.
- **Property Revenue Concentration:** Analysis revealed that **premium properties (top-tier listings) contributed nearly 60–70% of total sales value**, indicating a strong concentration of revenue in high-value assets.
- **Location-Based Insights:** The dashboard highlighted that **urban and high-demand locations generated higher sales and faster property turnover**, while suburban areas showed slower growth patterns.
- **AI-Driven Forecasting:** Using Power BI forecasting and AI assistance, it was observed that **property sales are expected to increase in upcoming quarters**, especially in high-growth zones, helping in future investment planning.
- **Customer Segmentation:** AI-based clustering categorized buyers into segments such as **first-time buyers, investors, and premium clients**, enabling targeted marketing and sales strategies.
- **Anomaly Detection in Sales Trends:** The system identified **irregularities such as sudden price drops or unexpected spikes in property transactions**, which may indicate market fluctuations or external economic factors.
- **Time Efficiency through AI Tools:** The use of AI tools reduced the time required for generating reports and insights from **hours to a matter of minutes**, enhancing productivity during the project.
- **Improved Decision-Making:** AI-generated narratives and predictive insights enabled **data-driven decision-making** for property portfolio management and investment strategies.
- **Enhanced Visualization and User Experience:** The Power BI dashboard provided an **interactive and user-friendly interface**, allowing stakeholders to easily analyze property performance, sales trends, and regional comparisons.
- **Reduction in Manual Effort:** Automation through AI significantly reduced manual analysis, minimizing human error and improving overall efficiency.

5.2 Project Links

Share the relevant links to your project work for evaluation.

- GitHub Link of the Project: https://github.com/rajviba934-art/Real-Estate-dashboard-.git
- Power BI / Dashboard Link (if applicable): [Paste published dashboard link here]

5.3 Limitations

Discuss the limitations of the current project or approach honestly.

- **Data Latency:** The system utilizes a static dataset; real-time data streaming was not implemented.
- **Sample Size:** The dataset represents a sample and may not account for global macroeconomic shifts.

5.4 Future Work

- **Predictive Analytics:** Integrating Machine Learning for price forecasting and trend prediction.
- **Live Integration:** Establishing API connections to live property listing sites.

5.5 Conclusion

The **Real Estate Sales and Property Portfolio Dashboard** project successfully addressed the challenge of managing disorganized property transaction data, which often leads to inefficient analysis and poor investment decisions in the real estate sector. By utilizing **Microsoft Excel** for rigorous data cleaning and **Power BI** for creating an interactive dashboard, I transformed raw transaction records into a centralized, visual analytical system. The analysis provided critical insights by identifying **high-demand geographic locations**, revealing the **dominance of residential property sales**, and highlighting **seasonal trends** and pricing patterns.

This project adds significant value by providing firms with a structured tool to monitor **Total Sales Value**, **Inventory Status**, and **Average Property Prices**, effectively moving from manual reporting to automated business intelligence. All project objectives—from data collection and preprocessing to **AI integration with Microsoft Copilot and ChatGPT** for insight generation—were fully achieved. Personally, this fellowship experience has been invaluable, greatly enhancing my practical skills in **Data Analytics**, **interactive visualization**, and the application of **AI-assisted reporting** in a real-world domain.

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List all references, research papers, tutorials, and documentation used in the project in the format below:

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APPENDICES

Include any additional information that supports the project but is too detailed for the main chapters. This may include:

Appendix A — Sample Dataset

Property ID	Location	Property Type	Property Price	Area Size (sq ft)	Sales Date	Status
101	Mumbai	Apartment	7,500,000	1200	12-01-2024	Sold
102	Delhi	Villa	12,000,000	2000	18-02-2024	Available
103	Delhi	Apartment	8,200,000	1400	05-03-2024	Sold
104	Ahmedabad	House	6,500,000	1500	22-03-2024	Available
105	Mumbai	Villa	11,000,000	1800	14-04-2024	

Appendix B — Key Excel Formulas / DAX Measures Used

- **Excel Formulas:**
  - =SUM(Property_Price_Range) — Total Sales Calculation.
  - =AVERAGE(Property_Price_Range) — Average Price metric.
  - =COUNTIF(Status_Range, "Sold") — Total Properties Sold.
  - =COUNTIF(Status_Range, "Available") — Total Properties Available.
- **Power BI DAX Measures:**
  - Total Sales = SUM(Properties[Property Price])
  - Sold Properties = CALCULATE(COUNT(Properties[Property ID]), Properties[Status] = "Sold")

Appendix C — AI Prompts Used

1. "Analyze this real estate dataset and identify the top-performing locations based on property sales."
2. "Acting as a market researcher, summarize the trends observed in the monthly property sales data."
3. "Generate a professional executive summary for a Power BI dashboard showing real estate sales analysis."

Appendix D — Screenshot Gallery

Figure D1: *Importing Raw Dataset into Excel*

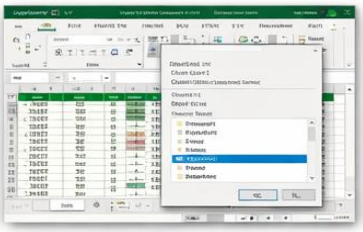

Figure D2: *Identifying Data Inconsistencies using Filters*

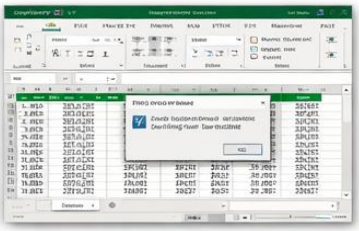

Figure D3: *Removing Duplicate Records*

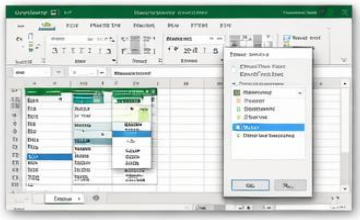

Figure D4: *Handling Missing Data Values*

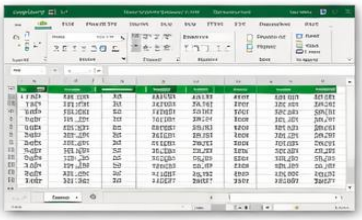

Figure D5: *Standardizing Data Formats*

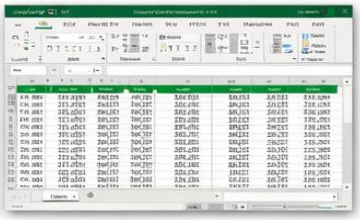

Figure D6: *Cleaned and Structured Dataset*

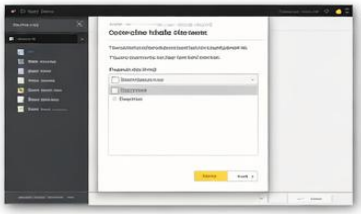

Figure D7: *Importing Data into Power BI*

Figure D8: *Data Model and Relationships*

Figure D9: *Dashboard Development in Power BI*

Figure D10: *Interactive Filters and Slicers*

Figure D11: *Drill-Down Analysis*

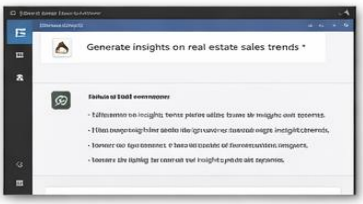

Figure D12: *AI Prompt Input*

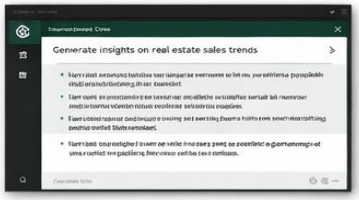

Figure D14: *Ai-Generated Analytical Insights*

Figure D15: *Integration of AI Insights*

Figure D16: *Additional Data Visualizations*